EE3110 - Probability Foundations for Electrical Engineers
Tutorial - 2

1. Suppose a test for COVID is 99% accurate in both directions (positive and negative) and 0.3% of the population is COVID positive. If someone tests positive, what is the probability they actually are COVID positive? Justify why you are getting a low probability even though the test appears to be very accurate.

Solution: Let D is the event that a person is COVID positive, and T is the event that the person tests positive.

$$\mathbb{P}(D \mid T) = \frac{\mathbb{P}(D \cap T)}{\mathbb{P}(T)} = \frac{(0.99)(0.003)}{(0.99)(0.003) + (0.01)(0.997)} \approx 23\%$$

A short reason why this surprising result holds is that the error in the test is much greater than the percentage of people with COVID. This leads to a lot of false positives than "expected".

2. We throw 3 dice one by one. What is the probability that we obtain 3 numbers in strictly increasing order?

Solution: To have 3 numbers in strictly increasing order, first all three numbers must be different numbers. Conditioned on three different numbers, the probability of strictly increasing order is simply $1/3! = 1/6$ (one specific sequence out of all possible permutations).

Thus,

$$\begin{aligned}\mathbb{P}(\text{increasing}) &= \mathbb{P}(\text{increasing } \cap \text{ all different}) \\ &= \mathbb{P}(\text{all different}) \times \mathbb{P}(\text{increasing} \mid \text{all different}) \\ &= \frac{6 \cdot 5 \cdot 4}{6^3} \times \frac{1}{6} = \frac{5}{9} \times \frac{1}{6} = \boxed{\frac{5}{54}}.\end{aligned}$$

3. A coin is tossed twice. Harshith claims that the event of two heads is at least as likely if we know that the first toss is a head than if we know that at least one of the tosses is a head. Is he right? Does it make a difference if the coin is fair or unfair? How can we generalize Harshith's reasoning?

Solution: Let A be the event that the first toss is a head and B be the event that second toss is a head. Harshith claims that $\mathbb{P}(A \cap B|A) \geq \mathbb{P}(A \cap B|A \cup B)$. Let us analyze the claim:

$$\mathbb{P}(A \cap B|A) = \frac{\mathbb{P}(A \cap B \cap A)}{\mathbb{P}(A)} = \frac{\mathbb{P}(A \cap B)}{\mathbb{P}(A)}. \tag{1}$$

$$\mathbb{P}(A \cap B|A \cup B) = \frac{\mathbb{P}((A \cap B) \cap (A \cup B))}{\mathbb{P}(A \cup B)} = \frac{\mathbb{P}(A \cap B)}{\mathbb{P}(A \cup B)}. \tag{2}$$

Since, $\mathbb{P}(A \cup B) \geq \mathbb{P}(A)$, we have **(1)** greater than **(2)** and hence Harshith is correct irrespective of the coin being fair or not.
A generalization of Harshith's reasoning is that if A', B', and C' are events such that $B' \subset C'$ and $A' \cap B' = A' \cap C'$ then the event A' is at least as likely if we know that B' has occurred; than if we know that C' has occurred.

4. Let A and B be events with $\mathbb{P}(A) > 0$ and $\mathbb{P}(B) > 0$. We say that an event B *suggests* an event A if $\mathbb{P}(A|B) > \mathbb{P}(A)$, and does not suggest event A if $\mathbb{P}(A|B) < \mathbb{P}(A)$.

   1. Show that B suggests A if and only if A suggests B.
   2. Show that B suggests A if and only if B^c does not suggest A. Assume that $\mathbb{P}(B^c) > 0$.
   3. We know that a treasure is located in one of two places, with probabilities β and $1-\beta$, respectively, where $0 < \beta < 1$. We search the first place and if the treasure is there, we find it with probability $p > 0$. Show that the event of not finding the treasure in the first place suggests that the treasure is in the second place.

Solution: 1. We have:

$$P(B|A) = \frac{P(A|B)\,P(B)}{P(A)} > P(B)$$

Since the probabilities are assumed to be non-zero, the above equation can be re-written as:

$$P(A|B) > P(A)$$

which is exactly the definition of B suggesting A. Hence, the forward implication is proved. A similar proof can be used for the backward implication.

2. Aim: To show B suggests A if and only if B^c does not suggest A.

If B^c does not suggest A, then B suggests A. We have:

$$P(A|B^c) = \frac{P(B^c|A)\,P(A)}{P(B^c)} < P(A)$$

From this equation, we have:

$$P(B^c|A) = 1 - P(B|A) < P(B^c) = 1 - P(B)$$

$$P(B|A) > P(B)$$

Hence, we've shown A suggests B. Using (1), we can tell that B suggests A. Hence, we've proved the forward implication. The same proof can be started from the end and reversed to prove the backward implication.

3. Given:
P(Treasure is in first place) = β, P(Treasure is in second place) = $1 - \beta$.
P(Finding treasure given treasure is present) = p

Event A: Not finding the treasure in first place
Event B: Treasure is in the second place

Aim: To show that A suggests B. Event A can occur in two cases:

1. Treasure is not present at the first place (or)
2. Treasure is present but it was not found

Clearly, we can tell that event B is a proper subset of event A (Event that treasure is in the second place means it is not in the first place which is part of the event that treasure is not found in the first place).
It is a proper subset since event A can happen even if event B is not true (First place contains the treasure but it was not found)

Thus, we have $B \subset A$. Hence, $P(A|B) = \frac{P(A \cap B)}{P(B)} = \frac{P(B)}{P(B)} = 1 > P(A)$. We've shown that B suggests A. Using (1), we can show A too suggests B.

5. For events A, B and C with $C \subseteq B$ and $P(A)P(B)P(C) > 0$, $P(A|B) = \alpha P(A|B \cap C) + (1-\alpha)P(A|B \cap C^c)$. Find α in terms of $P(A), P(B)$ and $P(C)$.

Solution:

Using total probability theorem, we know that,

$$A = (A \cap C) \cup (A \cap C^c)$$

Note that by the definition of P_B and the fact that it is indeed another probability map (i.e., satisfies the axioms of probability), we have,

$$\begin{aligned} P(A|B) = P_B(A) &= P_B((A \cap C) \cup (A \cap C^c)) \\ &= P_B(A \cap C) + P_B(A \cap C^c) \\ &= P_B(C)P_B(A|C) + P_B(C^c)P_B(A|C^c) \end{aligned}$$

By the fact that

$$P_B(A|C) = P(A|B \cap C)$$

we have,

$$P(A|B) = P(C|B)P(A|B \cap C) + P(C^c|B)P(A|B \cap C^c)$$

Comparing with the expression given in the question,

$$\alpha = P(C|B) = \tfrac{P(C)}{P(B)} \text{ (as C is a subset of B).}$$

6. A large class in probability theory is taking a multiple-choice test. For a particular question on the test, the fraction of examinees who know the answer is p; $1-p$ is the fraction that will guess. The probability of answering a question correctly is unity for an examinee who knows the answer and $1/m$ for a guesser; m is the number of multiple choice alternatives. Compute the probability that an examinee knew the answer to a question, given that he or she has answered it correctly.

Solution: Let K be the event that an examinee knows the answer, G the event of the examinee guesses, and C the event that the answer is correct.

$$\mathbb{P}(K) = p, \;\; \mathbb{P}(G) = 1 - p$$

$$\mathbb{P}(C|K) = 1, \;\; \mathbb{P}(C|G) = 1/m$$

We need to find

$$\begin{aligned} \mathbb{P}(K|C) &= \frac{\mathbb{P}(C|K)\mathbb{P}(K)}{\mathbb{P}(C|K)\mathbb{P}(K) + \mathbb{P}(C|G)\mathbb{P}(G)} \\ &= \frac{p}{p + (1-p)/m} \end{aligned}$$

7. *The prisoner's dilemma*: The release of two out of three prisoners has been announced, but their identity is kept secret. One of the prisoners considers asking a friendly guard to tell him who is the prisoner other than himself that will be released, but hesitates based on the following rationale: at the prisoner's

present state of knowledge, the probability of being released is $\frac{2}{3}$, but after he knows the answer, the probability of being released will become $\frac{1}{2}$, since there will be two prisoners (including himself) whose fate is unknown and exactly one of the two will be released. What is wrong with this line of reasoning?

Solution: Intuitively, there is something wrong with this rationale. The reason is that it is not based on a correctly specified probabilistic model. In particular, the event where both of the other prisoners are to be released is not properly accounted in the calculation of the posterior probability of release.
To be precise, let A, B, and C be the prisoners, and let A be the one who considers asking the guard. Suppose that all prisoners are a priori equally likely to be released. Suppose also that if B and C are to be released, then the guard chooses B or C with equal probability to reveal to A. Then, there are four possible outcomes:

1. A and B are to be released, and the guard says B (probability 1/3)
2. A and C are to be released, and the guard says C (probability 1/3)
3. B and C are to be released, and the guard says B (probability 1/6)
4. B and C are to be released, and the guard says C (probability 1/6)

Thus, P(A is to be released | guard says B) = P(A is to be released and guard says B)/P(guard says B) = $\frac{\frac{1}{3}}{\frac{1}{3}+\frac{1}{6}} = \frac{2}{3}$
Similarly, P(A is to be released | guard says C) = $\frac{2}{3}$
Thus, regardless of the identity revealed by the guard, the probability that A is released is equal to 2/3, the a priori probability of being released.

8. **(a)** Consider the following 7-door version of the Monty Hall problem. There are 7 doors, behind one of which there is a car (which you want), and behind the rest of which there are goats (which you don't want). Initially, all possibilities are equally likely for where the car is. You choose a door. Monty Hall then opens 3 goat doors, and offers you the option of switching to any of the remaining 3 doors.

   Assume that Monty Hall knows which door has the car, will always open 3 goat doors and offer the option of switching, and that Monty chooses with equal probabilities from all his choices of which goat doors to open. Should you switch? What is your probability of success if you switch to one of the remaining 3 doors?

   (b) Generalize the above to a Monty Hall problem where there are $n \geq 3$ doors, of which Monty opens m goat doors, with $1 \leq m \leq n-2$.

Solution:

(a) Assume the doors are labeled such that you choose door 1 (to simplify notation), and suppose first that you follow the "stick to your original choice" strategy. Let S be the event of success in getting the car, and let C_j be the event that the car is behind door j. Conditioning on which door has the car, we have

$$P(S) = P(S|C_1)P(C_1) + \cdots + P(S|C_7)P(C_7) = P(C_1) = \frac{1}{7}.$$

Let M_{ijk} be the event that Monty opens doors i, j, k. Then

$$P(S) = \sum_{i,j,k} P(S|M_{ijk})P(M_{ijk})$$

(summed over all i, j, k with $2 \le i < j < k \le 7$.) By symmetry, this gives

$$P(S|M_{ijk}) = P(S) = \frac{1}{7}$$

for all i, j, k with $2 \le i < j < k \le 7$. Thus, the conditional probability that the car is behind 1 of the remaining 3 doors is 6/7, which gives 2/7 for each. So you should switch, thus making your probability of success 2/7 rather than 1/7.

(b) By the same reasoning, the probability of success for "stick to your original choice" is $\frac{1}{n}$, both unconditionally and conditionally. Each of the $n-m-1$ remaining doors has conditional probability

$$\frac{n-1}{(n-m-1)n}$$

of having the car. This value is greater than $\frac{1}{n}$, so you should switch, thus obtaining probability

$$\frac{n-1}{(n-m-1)n}$$

of success (both conditionally and unconditionally).

9. To encourage Elmer's promising tennis career, his father offers him a prize if he wins (at least) two tennis sets in a row in a three-set series to be played with his father and the club champion alternately: father–champion–father or champion–father–champion, according to Elmer's choice. The champion is a better player than Elmer's father. Which series should Elmer choose?

Solution: Since the champion plays better than the father, it seems reasonable that fewer sets should be played with the champion. On the other hand, the middle set is the key one, because Elmer cannot have two wins in a row without winning the middle one. Let C stand for champion, F for father, and W and L for a win and a loss by Elmer. Let f be the probability of Elmer's winning any set from his father, c the corresponding probability of winning from the champion. The table shows the only possible prize-winning sequences together with their probabilities, given independence between sets, for the two choices.

	Father first				**Champion first**			
Set with:	F	C	F	Probability	C	F	C	Probability
	W	W	W	fcf	W	W	W	cfc
	W	W	L	$fc(1-f)$	W	W	L	$cf(1-c)$
	L	W	W	$(1-f)cf$	L	W	W	$(1-c)fc$
Totals				$fc(2-f)$				$fc(2-c)$

Since Elmer is more likely to best his father than to best the champion, f is larger than c, and $2-f$ is smaller than $2-c$, and so Elmer should choose CFC. For example, for $f = 0.8$, $c = 0.4$, the chance of winning the prize with FCF is 0.384, that for CFC is 0.512. Thus the importance of winning the middle game outweighs the disadvantage of playing the champion twice.

Many of us have a tendency to suppose that the higher the expected number of successes, the higher the probability of winning a prize, and often this supposition is useful. But occasionally a problem has special conditions that destroy this reasoning by analogy. In our problem the expected number of wins under CFC is $2c + f$, which is less than the expected number of wins for FCF, $2f + c$. In our example with $f = 0.8$ and $c = 0.4$, these means are 1.6 and 2.0 in that order. This opposition of answers gives the problem its flavor.

10. Alice searches for her term paper in her filing cabinet, which has several drawers. She knows that she left her term paper in drawer j with probability $p_j > 0$. The drawers are so messy that even if the paper is present in drawer i, the probability that she finds it is only d_i.

    Alice unsuccessfully searches in a particular drawer i. Conditioned on this event, find the probability that the paper is in drawer j for: (a) $j \neq i$, (b) $j = i$.

Solution: Let A be the event that Alice doesn't find her paper in drawer i and B be the event that the paper is in drawer j.

$$\mathbf{P}\left(A^c\right) = p_i d_i \qquad \mathbf{P}(A) = 1 - p_i d_i \qquad \mathbf{P}(B) = p_j$$

(a) $j \neq i$

$\mathbf{P}(A \cap B)$ = Probability of paper in drawer j and can't be found in drawer i = Probability of paper in drawer $j = \mathbf{P}(B)$

$$\mathbf{P}(B \mid A) = \frac{\mathbf{P}(A \cap B)}{\mathbf{P}(A)} = \frac{\mathbf{P}(B)}{\mathbf{P}(A)} = \frac{p_j}{1 - p_i d_i}$$

(b) $i = j$

$\mathbf{P}(A \cap B)$ = Probability of paper in drawer j and can't be found in drawer $j = \mathbf{P}(B)\mathbf{P}(A \mid B)$ = Probability of paper being in drawer j * Probability of not finding the paper even if it is present in drawer j

$$\mathbf{P}(B \mid A) = \frac{\mathbf{P}(A \cap B)}{\mathbf{P}(A)} = \frac{\mathbf{P}(B)\mathbf{P}(A \mid B)}{\mathbf{P}(A)} = \frac{p_i\left(1 - d_i\right)}{1 - p_i d_i}$$

11. You are taking out candies one by one from a jar that has 10 red candies, 20 blue candies, and 30 green candies in it. What is the probability that there are at least 1 blue candy and 1 green candy left in the jar when you have taken out all the red candies?

Solution: At first look, this problem appears to be a combinatorial one. However, a conditional probability approach gives a much more intuitive answer. Let T_r, T_b, and T_g be the number that the last red, blue, and green candies are taken out respectively.

To have at least 1 blue candy and 1 green candy left when all the red candies are taken out, we need to have $T_r < T_b$ and $T_r < T_g$. In other words, we want to derive:

$$P(T_r < T_b \cap T_r < T_g)$$

There are two mutually exclusive events that satisfy $T_r < T_b$ and $T_r < T_g$:

$$T_r < T_b; \quad T_r < T_g < T_b$$

and

$$T_r < T_g; \quad T_r < T_b < T_g$$

Therefore,

$$P(T_r < T_b \cap T_r < T_g) = P(T_r < T_b < T_g) + P(T_r < T_g < T_b)$$

Case 1: $T_r < T_b < T_g$ means that the last candy is green ($T_g = 60$). Since each of the 60 candies are equally likely to be the last candy and among them 30 are green ones, we have:

$$P(T_g = 60) = \frac{30}{60}$$

Conditioned on $T_g = 60$, we need $P(T_r < T_b \mid T_g = 60)$. Among the 30 red and blue candies, each candy is equally likely to be the last candy and there are 20 blue candies, so:

$$P(T_r < T_b \mid T_g = 60) = \frac{20}{30}$$

Thus,

$$P(T_r < T_b < T_g) = \frac{30}{60} \times \frac{20}{30}$$

Case 2: Similarly, $T_r < T_g < T_b$ means that the last candy is blue ($T_b = 60$). Since each of the 60 candies are equally likely to be the last candy and among them 20 are blue ones, we have:

$$P(T_b = 60) = \frac{20}{60}$$

Conditioned on $T_b = 60$, we need $P(T_r < T_g \mid T_b = 60)$. Among the 40 red and green candies, each candy is equally likely to be the last candy and there are 30 green candies, so:

$$P(T_r < T_g \mid T_b = 60) = \frac{30}{40}$$

Thus,

$$P(T_r < T_g < T_b) = \frac{20}{60} \times \frac{30}{40}$$

Final probability:

$$P(T_r < T_b \cap T_r < T_g) = \frac{30}{60} \times \frac{20}{30} + \frac{20}{60} \times \frac{30}{40} = \frac{7}{12}$$

12. Consider a coin that comes up heads with probability p and tails with probability $1 - p$. Let q_n be the probability that after n independent tosses, there have been an even number of heads. Derive a recursion that relates q_n to q_{n-1}, and solve this recursion to establish the formula

$$q_n = \frac{1 + (1 - 2p)^n}{2}.$$

Solution:

Let A be the event of getting an even number of heads in $n - 1$ tosses, and let B be the event that the nth toss is heads. For having an even number of heads in n tosses,

- There need to be an odd number of heads in $n - 1$ tosses followed by a heads, (OR),

- There need to be an even number of heads in $n-1$ tosses followed by a tails.

Therefore, q_n can be written as follows:

$$\begin{aligned} q_n &= \mathbb{P}\big((A^c \cap B) \cup (A \cap B^c)\big) \\ &= \mathbb{P}(A^c)\,\mathbb{P}(B) + \mathbb{P}(A)\,\mathbb{P}(B^c) \\ &= (1 - q_{n-1})\,p \;+\; q_{n-1}\,(1-p) \\ &= (1-2p)\,q_{n-1} \;+\; p \end{aligned}$$

Now, we solve this by induction. It is easy to verify this for the case $n = 1$, i.e. $q_1 = 1 - p$. Next, we inductively assume that $q_{n-1} = \frac{1+(1-2p)^{n-1}}{2}$. Substituting this in the above expression, we get

$$q_n = \boxed{\frac{(1+(1-2p)^n)}{2}},$$

which proves the inductive hypothesis.

13. A line of 100 airline passengers is waiting to board a plane. They each hold a ticket to one of the 100 seats on that flight. For convenience, let's say that the n-th passenger in line has a ticket for seat number n. Being drunk, the first person in line picks a random seat (equally likely for each seat). All of the other passengers are sober, and will go to their assigned seats unless it is already occupied; If it is occupied, they will then find a free seat to sit in, at random. What is the probability that the last (100th) person to board the plane will sit in their own seat?

Solution: Note that the last passenger can only take seat #1 or #100.

- If any passenger takes seat #1, the cycle stops, and all the subsequent passengers take their own seats (including the last)
- Otherwise, if the #100 seat is taken before #1, the cycle is paused, i.e., the subsequent passengers do take their own seats, but the last passenger would take seat #1.

Now, any passenger from 1st to 99th who is picking a random vacant seat will choose between #1, #100 or any other seat equally likely. Thus, by symmetry, #1 or #100, anyone will be taken first - with equal probability.

Hence the last person ends up in their own seat with a probability of 0.5.

14. There are N urns of which the r-th urn contains $r-1$ red balls and $N-r$ blue balls. You pick an urn at random and remove two balls at random without replacement. Find the probability that
    (a) the second ball is blue.
    (b) the se cond ball is blue, given that the first is blue.

Solution: The rth urn contains $r-1$ red and $N-r$ blue balls. Hence the total number of balls $= N-1$.

(a)

$$\begin{aligned}P(\text{2nd ball is blue} \mid r\text{th urn is chosen}) &= P(\text{2nd is blue} \mid r\text{th urn, 1st is red}) \cdot P(\text{1st is red} \mid r\text{th urn}) \\ &\quad + P(\text{2nd is blue} \mid r\text{th urn, 1st is blue}) \cdot P(\text{1st is blue} \mid r\text{th urn}) \\ &= \frac{N-r}{N-2} \cdot \frac{r-1}{N-1} + \frac{N-r-1}{N-2} \cdot \frac{N-r}{N-1} \\ &= \frac{N-r}{N-1}\end{aligned}$$

$$\begin{aligned}P(\text{second ball is blue}) &= \sum_{r=1}^{N} P(\text{second ball is blue} \mid r\text{th urn is selected}) \cdot P(r\text{th urn is selected}) \\ &= \sum_{r=1}^{N} \frac{N-r}{N-1} \cdot \frac{1}{N} = \frac{1}{2}\end{aligned}$$

(b)

$$P(\text{second ball is blue} \mid \text{first ball is blue}) = \frac{P(\text{both the balls are blue})}{P(\text{first ball is blue})}$$

$$\begin{aligned}P(\text{first ball is blue}) &= \sum_{r=1}^{N} P(\text{first ball is blue} \mid r\text{th urn is selected}) \cdot P(r\text{th urn is selected}) \\ &= \sum_{r=1}^{N} \frac{N-r}{N-1} \cdot \frac{1}{N} = \frac{1}{2}\end{aligned}$$

$$P(\text{both the balls blue} \mid r\text{th urn is selected}) = \begin{cases} \frac{(N-r)(N-r-1)}{(N-1)(N-2)}, & r = 1, \ldots, N-2, \\ 0, & r = N \end{cases}$$

$$\begin{aligned}P(\text{both the balls blue}) &= \sum_{r=1}^{N-2} \frac{(N-r)(N-r-1)}{(N-1)(N-2)} \cdot \frac{1}{N} \\ &= \frac{1}{N(N-1)(N-2)} \sum_{r=1}^{N-2} (N-r)(N-r-1) \\ &= \frac{1}{3}\end{aligned}$$

$$\begin{aligned}P(\text{second ball is blue} \mid \text{first ball is blue}) &= \frac{P(\text{both the balls are blue})}{P(\text{first ball is blue})} \\ &= \frac{\frac{1}{3}}{\frac{1}{2}} = \frac{2}{3}\end{aligned}$$

15. **Gambler's Ruin.** Suppose you play a game by tossing a fair coin repeatedly and independently. If it

comes up heads, you win a dollar, and if it comes up tails, you lose a dollar. Suppose you start with \$50. What is the probability you will get to \$200 without first getting ruined (running out of money)?

Solution: For any given amount n of current holdings, the conditional probability of reaching N dollars before going broke is independent of how we acquired the n dollars, so there is a unique probability $\mathbb{P}(N \mid n)$ of reaching N on the condition that we currently hold n dollars. For any finite N we see that $\mathbb{P}(N \mid n) = 0$ and $\mathbb{P}(N \mid N) = 1$. We are considering this setting for $N = 200$, and we would like to find $\mathbb{P}(200 \mid 50)$. Denote $y(n) := \mathbb{P}(200 \mid n)$, which is the probability you get to 200 without first getting ruined if you start with n dollars. We saw that $y(0) = 0$ and $y(200) = 1$. Suppose the player has n dollars at the moment, the next round will leave the player with either $n + 1$ or $n - 1$ dollars, both with probability $1/2$. Thus the current probability of winning is the same as the weighted average of the probabilities of winning in the player's two possible next states. So we have

$$y(n) = \tfrac{1}{2}y(n+1) + \tfrac{1}{2}y(n-1).$$

Multiplying by 2, and subtracting $y(n) + y(n-1)$ from each side, we have

$$y(n+1) - y(n) = y(n) - y(n-1).$$

This says that the slopes of the graph of $y(n)$ on the adjacent intervals are constant. In other words, the graph of $y(n)$ must be a line. Since $y(0) = 0$ and $y(200) = 1$, we have $y(n) = n/200$, and therefore $y(50) = 1/4$.

Another way to see what the function $y(n)$ is to use the telescoping sum as follows

$$\begin{aligned} y(n) = y(n) - y(0) &= \big(y(n) - y(n-1)\big) + \cdots + \big(y(1) - y(0)\big) \\ &= n\big(y(1) - y(0)\big), \end{aligned}$$

since the differences are all the same, and $y(0) = 0$. To find $y(1)$ we can use the fact that $y(200) = 1$, so $y(1) = 1/200$, and therefore $y(n) = n/200$ and $y(50) = 1/4$.